

ALGEBRA BASICS



GRADE 6

What is Substitution?

Substitution is an algebraic technique used to solve equations and simplify expressions by replacing one or more variables (letters) with specific numbers.



***It's like 'subbing' a
player in a soccer or
basketball match!***

How do we substitute?

Step 1: Understand Variables

Step 2: Identify the Variable/s to Substitute

Step 3: Substitute and Simplify

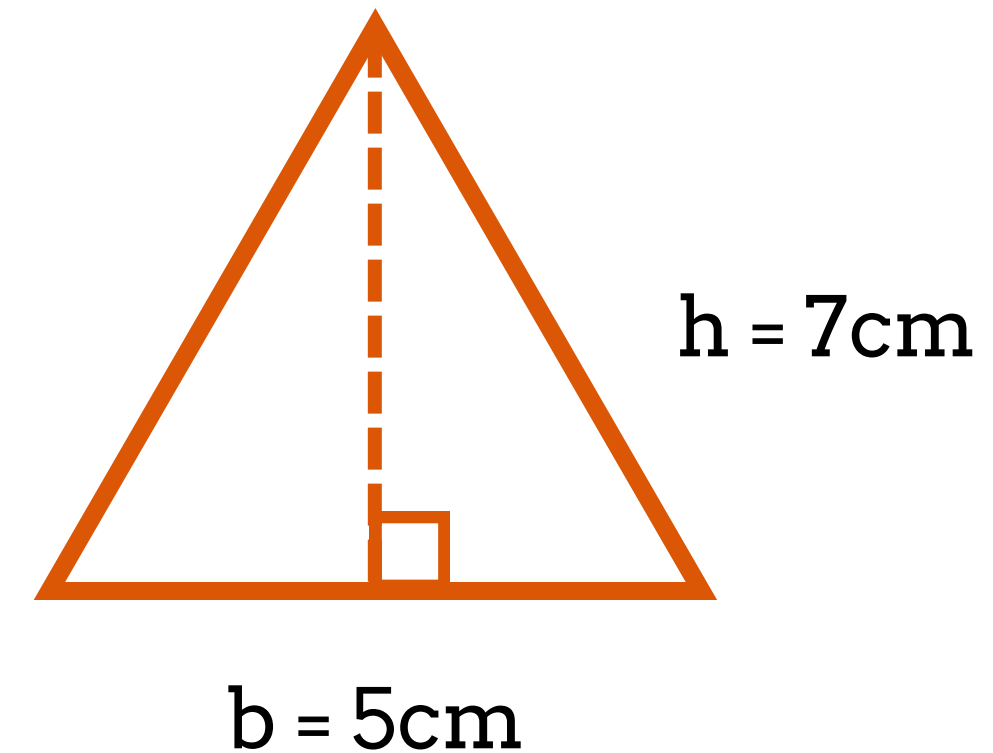


Let's Try

The formula for the area of a triangle is given by: **$a = bh/2$**

Step 1: Understand Variables

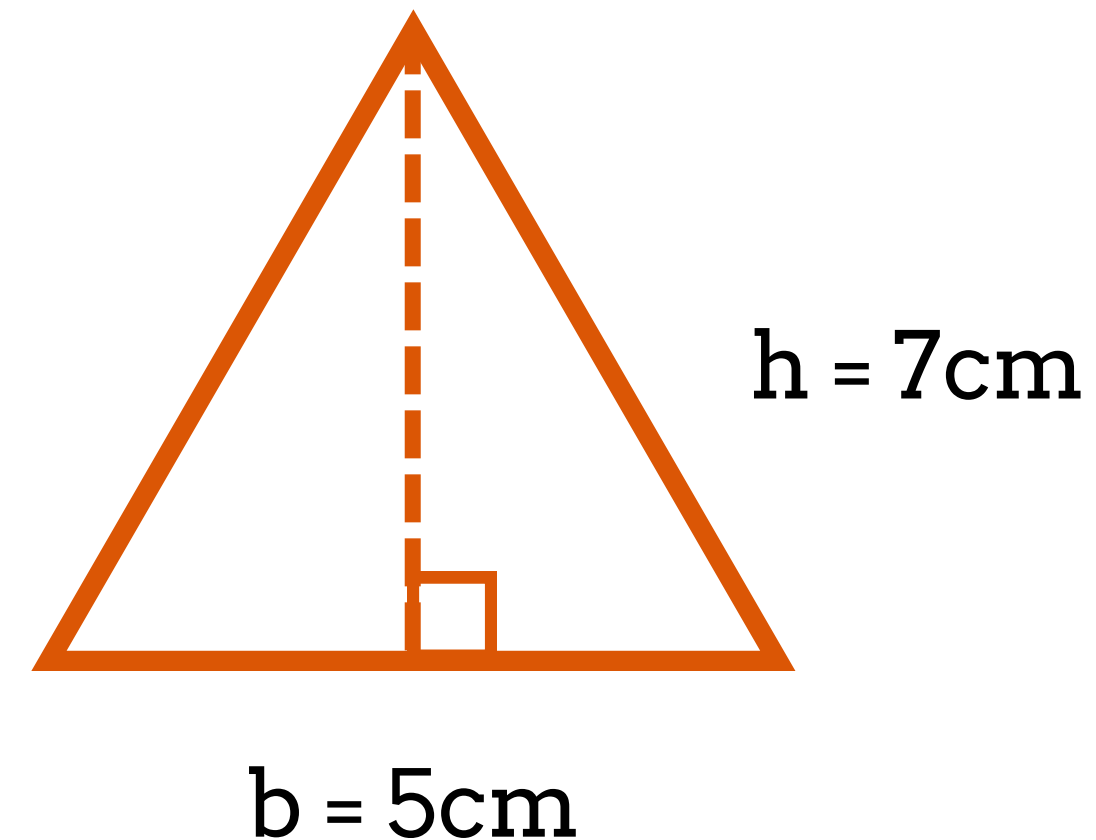
- a = area of the triangle
- b = base length of the triangle
- h = height of the triangle



Step 2: Identify the Variable to Substitute

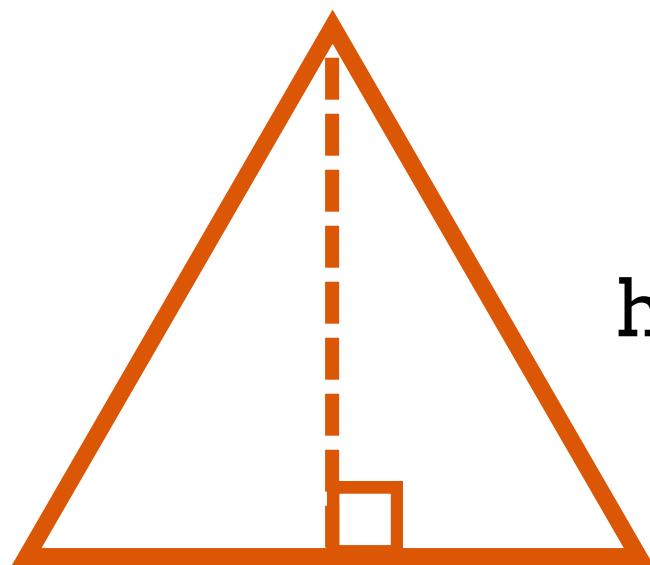
We can substitute values into b and h because these values have been given to us in the diagram.

We can use these values to find the area of the triangle (a).



Step 3: Substitute and Simplify

Substitute the given values into b and h to simplify and solve for a.



$h = 7\text{cm}$

$b = 5\text{cm}$

$a = \text{unknown}$

$h = 7\text{cm}$

$b = 5\text{cm}$

$a = bh/2$

$a = 5 \times 7 / 2$

$a = 35 / 2$

$a = 17.5\text{cm}^2$

The area of the triangle is 17.5cm^2 .



Here's another

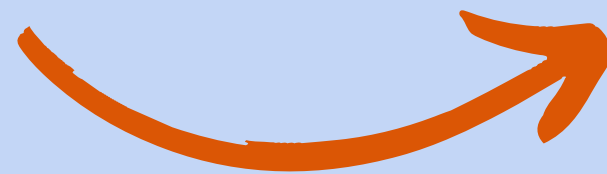
The formula for simple interest is given by: **$I = PRT$**

Step 1: Understand Variables

- I = amount of interest (\$)
- P = initial investment (\$)
- R = interest rate (as a decimal)
- T = time (years)

Step 2: Identify the Variable/s to Substitute

If we are investing \$500 in the bank for 3 years at an interest rate of 3.5%, how much interest will we receive?



$I = \text{unknown}$

$P = 500$

$R = 0.035$

$T = 3$

Step 3: Substitute and Simplify

$I = \text{unknown}$

$P = 500$

$R = 3.5$

$T = 3$



$I = PRT$

$I = 500 \times 0.035 \times 3$

$I = 52.5$

We would receive \$52.50 in interest.

QUESTION 1

LINEAR ROUND

What is the coefficient in the expression
 $6x^2 + 4$?

A

6

B

2

C

x

D

4

QUESTION 1

LINEAR ROUND

What is the coefficient in the expression
 $6x^2 + 4$?

ANSWER:

A

6

QUESTION 2

LINEAR ROUND

In the expression $4x + 6 - 2x$, which terms are like terms?

A

$4x$ and 6

B

$4x$ and $-2x$

C

6 and $-2x$

D

None

QUESTION 2

LINEAR ROUND

In the expression $4x + 6 - 2x$, which terms are like terms?

ANSWER:

B

$4x$ and $-2x$

QUESTION 3

LINEAR ROUND

Which of the following is a term in the expression $3y + 8 - 4y$?

A

y

B

3

C

-4

D

8

QUESTION 3

LINEAR ROUND

Which of the following is a term in the expression $3y + 8 - 4y$?

ANSWER:

D

8

QUESTION 4

LINEAR ROUND

Simplify the following expression:

$$4(x-2) + 5$$

A

$$4x + 7$$

B

$$4x - 7$$

C

$$4x - 3$$

D

$$4x + 3$$

QUESTION 4

LINEAR ROUND

Simplify the following expression:

$$4(x-2) + 5$$

ANSWER:

C

$$4x - 3$$

QUESTION 5

LINEAR ROUND

How many terms are in the expression $3(x + 2) - 4x + 1$ after it is simplified?

A

1

B

2

C

3

D

4

QUESTION 5

LINEAR ROUND

How many terms are in the expression
 $3(x + 2) - 4x + 1$ after it is simplified?

ANSWER:

B

2

QUESTION 6

SQUARE ROUND

Find the value of x:

$$3x + 2 = 11$$

A

2

B

3

C

4

D

5

QUESTION 6

SQUARE ROUND

Find the value of x:

$$3x + 2 = 11$$

ANSWER:

B

3

QUESTION 7

SQUARE ROUND

Which of the following equations does NOT have the solution $x = 3$?

A

$$2x + 1 = 7$$

B

$$3x = 9$$

C

$$x - 4 = -1$$

D

$$x - 6 = 3$$

QUESTION 7

SQUARE ROUND

Which of the following equations does
NOT have the solution $x = 3$?

ANSWER:

D

$$x - 6 = 3$$

QUESTION 8

SQUARE ROUND

Which of the following equations has NO solution?

A

$$4X = X + 3X$$

B

$$2X + 4 = 10$$

C

$$X - 2 = 5$$

D

$$3X - 5 = 3X + 2$$

QUESTION 8

SQUARE ROUND

Which of the following equations has NO solution?

ANSWER:

D

$$3X - 5 = 3X + 2$$

QUESTION 1

CUBE ROUND

Jamie has 3 more than twice the number of pens that Leah has.

If Leah has x pens, write the expression that shows how many pens Jamie has.

QUESTION 1

CUBE ROUND

Jamie has 3 more than twice the number of pens that Leah has. If Leah has x pens, write the expression that shows how many pens Jamie has.

ANSWER:

$$3 + 2x$$

QUESTION 2

CUBE ROUND

Solve the equation:
$$2(x - 4) + 3 = 13$$

QUESTION 2

CUBE ROUND

ANSWER:

$$2x - 8 + 3 = 13$$

$$2x - 5 = 13$$

$$2x = 18$$

$$x = 9$$

QUESTION 3

CUBE ROUND

Liam is 4 years less than twice Emma's age.

If Emma is x years old, write the expression that shows how old Liam is.

QUESTION 3

CUBE ROUND

Liam is 4 years less than twice Emma's age. If Emma is x years old, write the expression that shows how old Liam is.

ANSWER:

$$2x - 4$$

QUESTION 4

CUBE ROUND

A phone plan charges a \$15 monthly fee plus \$2 for every gigabyte of data used.

If x is the number of gigabytes, write the expression that shows the total cost of the plan.

QUESTION 4

CUBE ROUND

A phone plan charges a \$15 monthly fee plus \$2 for every gigabyte of data used.
If x is the number of gigabytes, write the expression that shows the total cost of the plan.

ANSWER:

$$2x + 15$$



THANK YOU